

GROMACS and VMD

PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

Day 1/5

Installing VMD

VMD is distributed by the University of Illinois.

You need to download the appropriate Linux binary from the official VMD website.

<https://www.ks.uiuc.edu>

VMD downloads require accepting the VMD license and providing registration information before the download is available

```
cd ~/Downloads
```

```
ls
```

```
tar -xzf <filename>.tar.gz
```

```
cd vmd-1.9.4
```

For the traditional VMD Linux package, the installation procedure uses the `configure` script to generate an installation script.
generate the Makefile

```
./configure
```

```
cd src
```

```
make install
```

```
vmd
```

Installing GROMACS

Linux/Ubuntu

Current GROMACS release is **2026.3**

official documentation recommends CMake-based installation

Check the system

`lsb_release -a`

Check compiler and CMake

`gcc --version` `sudo apt install gcc`

`g++ --version` `sudo apt install g++`

`cmake --version` `sudo apt install cmake`

(GROMACS 2026 requires **CMake 3.28 or newer**)

Install the required packages

`sudo apt update`

`sudo apt install build-essential cmake wget tar`

Installing GROMACS

Download Gromacs: `wget https://ftp.gromacs.org/gromacs/gromacs-2026.3.tar.gz`

Extract: `tar -xzf gromacs-2026.3.tar.gz`

`cd gromacs-2026.3`

Create a build directory - keep compiled files separated from source files

`mkdir build`

`cd build`

“quick and dirty” installation: build the required FFT library itself and download the regression tests

`cmake .. \`

`-DGMX_BUILD_OWN_FFTW=ON \`

`-DREGRESSIONTEST_DOWNLOAD=ON`

Installing GROMACS

Compile GROMACS

`make -j$(nproc) ; nproc - available CPU cores`

after compilation

`make check`

run

`sudo make install`

default GROMACS installed under `/usr/local/gromacs`

Activate GROMACS

`source /usr/local/gromacs/bin/GMXRC`

`gmx --version`

If you see the GROMACS version, the installation worked

Installing GROMACS

To automatically load GROMACS whenever you open a terminal, add this to your `.bashrc`

```
echo 'source /usr/local/gromacs/bin/GMXRC' >> ~/.bashrc  
source ~/.bashrc  
gmx --version
```

Check
`gmx help`

`gmx --version`

`gmx -h`

our basic CPU version of GROMACS is ready.